



Universidade do Minho

Bases de Dados

Licenciatura em Ciências da Computação

Ano Letivo de 2025/2026

2º Semestre

PL02

Theme: System Definition, Requirements Definition

Case Study

Mr. Ricardo works in the administrative department of a private medical clinic. Throughout his workday, he manages appointment bookings made in person, by phone, and through the clinic's website. For each booking, Mr. Ricardo systematically records the patient's name, user number, phone contact, chosen doctor, specialty, and the date and time of the appointment. Whenever it is a new patient, additional data such as address, date of birth, and email are also recorded. Furthermore, when the patient attends the appointment, Mr. Ricardo records whether the appointment was carried out or cancelled, the amount to be paid, the type of payment (cash, card, or transfer), and the name of the employee who received the payment. If the patient uses health insurance, the name of the insurer and the policy number are also recorded.

(...)

Based on the presented case study and considering the various steps of the database development methodology, it is intended that:

- a) Analyze the various aspects of the case study, justify and explain why a database system should be used.
- b) Develop a list of description requirements that allows the development of a conceptual schema for a database for the presented case.
- c) Using Chen notation, sketch a CONCEPTUAL schema for a database capable of accommodating the information revealed by the presented case.